

41_Costa_Rica

October 18, 2025

1 How have annual growth rates of exports of goods and services evolved relative to the share of exports in GDP in Costa Rica between 1962 and 2024?

1.1 Abstract

Using World Bank World Development Indicators (WDI), this study examines the evolution of Costa Rica's export performance between 1962 and 2024. The analysis compares two key indicators: the annual growth rate of exports of goods and services (%) and the share of exports of goods and services in GDP (%). Over this sixty-two-year period, the share of exports in GDP displayed moderate volatility but an overall slight upward trend, reflecting a gradual increase in Costa Rica's integration into the global economy. Meanwhile, annual export growth rates were highly volatile, with large fluctuations year to year, yet the overall average growth remained roughly stable. These patterns highlight the dynamic nature of Costa Rica's export sector, illustrating periods of rapid expansion, occasional contraction, and the steady structural role of exports in the national economy.

1.2 1. Question

How have annual growth rates of exports of goods and services evolved relative to the share of exports in GDP in Costa Rica between 1962 and 2024?

- **Exports growth proxy:** Exports of goods and services (annual % growth)
- **Exports share proxy:** Exports of goods and services (% of GDP)

1.3 2. Data

- **Source:** World Bank World Development Indicators (WDI)
- **Indicators:**
 - Exports of goods and services (annual % growth)
 - Exports of goods and services (% of GDP)
- **Coverage:** Costa Rica, 1962–2024
- **Notes:** National-level data only

1.4 3. Method

1. Filtered dataset for Costa Rica and selected the two export-related indicators.
2. **Extracted relevant columns:** Year, Indicator Name, and Value.
3. Pivoted the dataset to create a side-by-side chronological comparison of export growth and export share in GDP.

4. Produced a dual-line time series plot to visualize the relationship, volatility, and long-term trends of Costa Rica's export performance.

(Analysis is descriptive; no causal inference applied.)

1.5 4. Results

- **Exports of goods and services (% of GDP):** Showed moderate volatility throughout the period, with an overall slight upward trend, indicating a gradual increase in the importance of exports within the national economy.
- **Exports of goods and services (annual % growth):** Exhibited high volatility with frequent large fluctuations from year to year, yet the overall average remained relatively stable over the long term.
- **Comparison:** The two indicators demonstrate different patterns: the export share increased gradually with moderate swings, while annual growth rates fluctuated sharply around a consistent long-term average. This highlights the contrast between steady structural integration into global trade and the short-term instability of export performance.

(Figure 1. Costa Rica: Annual Export Growth vs. Export Share of GDP, 1962–2024)

(Table 1. Pivoted dataset summary)

1.6 5. Interpretation

- The gradual increase in export share relative to GDP indicates Costa Rica's long-term integration into the global economy, even amid periods of economic turbulence.
- High volatility in annual export growth reflects the sensitivity of Costa Rica's export sector to external shocks, commodity price swings, and global demand fluctuations.
- The differing patterns between growth rates and GDP share emphasize the distinction between structural export importance and year-to-year performance variability.
- These trends underscore the need for economic policies that both stabilize export performance and continue to expand the role of exports in national income.

1.7 6. Limitations

- National-level data may mask sectoral differences, such as concentration in agriculture, manufacturing, or services.
- WDI data rely on national reporting and modeled estimates, which may introduce uncertainty, especially in early years.
- The descriptive analysis does not isolate causal factors driving volatility or structural export growth, such as trade agreements, investment flows, or exchange rate changes.

1.8 7. Next Steps / Extensions

- Disaggregate export data by sector to analyze the contribution of key industries to overall export performance.
- Explore correlations with macroeconomic variables, such as foreign direct investment, exchange rates, or trade policies.
- Compare Costa Rica's trends with other Central American economies to evaluate regional patterns of trade integration.

- Conduct time-series decomposition to separate long-term trends from short-term fluctuations and identify structural shifts in export dynamics.

```
[1]: # How have annual growth rates of exports of goods and services evolved
      ↪ relative to the share of exports in GDP in Costa Rica between 1962 and 2024?

import pandas as pd
import matplotlib.pyplot as plt
import os

# Folders
data_raw_folder = "data_raw/"
data_clean_folder = "data_clean/"
figures_folder = "figures/"

# Load CSV
filename = "trade_cri_filtered.csv" # Filtered dataset with only relevant rows
df = pd.read_csv(os.path.join(data_raw_folder, filename))

# Keep only needed columns
df = df[["Year", "Indicator Name", "Value"]]

# Convert Year and Value to numeric, drop invalid rows
df["Year"] = pd.to_numeric(df["Year"], errors="coerce")
df["Value"] = pd.to_numeric(df["Value"], errors="coerce")
df = df.dropna(subset=["Year", "Value"])

# Pivot indicators into separate columns
df_pivot = df.pivot(index="Year", columns="Indicator Name", values="Value").
    ↪reset_index()
df_pivot = df_pivot.sort_values("Year")

print("Pivoted Costa Rica dataset:")
display(df_pivot)

# Interpolate missing values for smooth plotting (optional)
df_plot = df_pivot.interpolate(method='linear')

# Plot the indicators
plt.figure(figsize=(10,6))
plt.plot(df_plot["Year"], df_plot["Exports of goods and services (annual %
    ↪growth)"],
         marker='o', linestyle='-', label="Exports of goods and services
    ↪(annual % growth)")
plt.plot(df_plot["Year"], df_plot["Exports of goods and services (% of GDP)"],
         marker='o', linestyle='-', label="Exports of goods and services (% of
    ↪GDP)")
```

```

plt.title("Costa Rica: Exports of goods and services annual growth vs GDP_
↳percentage (%) (1962-2024)")
plt.xlabel("Year")
plt.ylabel("Percentage")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.savefig(os.path.join(figures_folder,
↳"costa_rica_exports_of_goods_and_services_annual_growth_vs_gdp_percentage.
↳png"))
plt.show()

# Save cleaned CSV
df_pivot.to_csv(os.path.join(data_clean_folder,
↳"costa_rica_exports_of_goods_and_services_annual_growth_vs_gdp_percentage"),
↳index=False)

```

Pivoted Costa Rica dataset:

Indicator Name	Year	Exports of goods and services (% of GDP) \
0	1962	22.723276
1	1963	22.087419
2	1964	24.419375
3	1965	22.563318
4	1966	24.780805
..	...	...
58	2020	31.906498
59	2021	36.176357
60	2022	42.405196
61	2023	38.915632
62	2024	38.531794

Indicator Name	Exports of goods and services (annual % growth)
0	11.404435
1	28.950575
2	-10.280374
3	2.200375
4	21.701787
..	...
58	-10.632856
59	15.933922
60	18.500877
61	9.959314
62	5.770045

[63 rows x 3 columns]



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